

Lecture 3 Part 2:

Hypothesis & Ethical Research

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The Hypothesis

What is a hypothesis?

Hypothesis: a **specific** and **testable** explanation for a phenomena.

- *Specific*: explains **what you expect to happen** in a certain circumstance;
- *Testable*: is **falsifiable** (can be disproven by empirical data);
- *Parsimonious*: is as **simple** as feasible (remember **Occam's Razor**);
- *Predictive*: makes a **prediction** that can be tested.

What is a hypothesis?

A hypothesis is an attempt to **explain an observation** that cannot be adequately described by existing accepted scientific theories.

- An informed guess about the **cause** of the phenomena.

Importantly, a hypothesis also allows **predictions** about unobserved entities.

These predictions are **tested** against empirical data from experiments.

What is a hypothesis?

Hypotheses and Theories are not the same:

- A scientific **hypothesis** is a testable prediction;
- A scientific **theory** is a hypothesis **proven** by testing;
 - i.e., the predictions were empirically validated.

- Remember from Part 1 Lecture 3: we might accept a hypothesis as a proven theory, but future evidence can falsify it
 - We can use new evidence to modify our disproven hypotheses
- NOTE: The scientific method is iterative

From hypothesis to experiment: example

From Hypothesis to Experiment

Imagine you've selected a problem and want to create an experiment. The the first things you need to consider are:

- What information can we gather to support the hypothesis? (the testability of the hypothesis)
- What would I need to discover to prove the hypothesis wrong? (the falsifiability of the hypothesis)

From Hypothesis to Experiment: Example

- Suppose we develop a new LLM algorithm “GlasgowGPT”
- Hypothesis: “GlasgowGPT is better than any other LLM algorithm”
- But this isn’t a very good hypothesis — what does “better” mean
 - Because “better” isn’t measurable

From Hypothesis to Experiment: Example

So we need consider:

- What do we mean by “better”
- What are these “other LLM algorithms”

From Hypothesis to Experiment: Example

So instead let's say:

- That “better” means it “gives more accurate responses”
- And these “other LLM algorithms” are the following algorithms:
“Claude, ChatGPT, and Grok”

From Hypothesis to Experiment: Example

Next:

- We need give each LLM a prompt which we can verify the facts from
 - For example: What is the capital of Egypt? Is the earth round?

From Hypothesis to Experiment: Example

Also need to ensure the competition is fair by controlling variables:

- Use the same type of LLM account (paid vs unpaid?)
- Using an account with no history
- Question order, and standard replies
- Use the same latest version
- Etc.

From Hypothesis to Experiment: Example

Now let's jump ahead and say “we've got our data, done the analysis, and want to make conclusions on our hypothesis”

Hypothesis: “GlasgowGPT is more accurate than Claude, ChatGPT and Grok when answering factual questions”

From Hypothesis to Experiment: Example

Hypothesis: “GlasgowGPT is more accurate than Claude, ChatGPT and Grok when answering factual questions”

- Why can't we find just one instance where this isn't true and conclude the hypothesis is false?
- There might be some obscure / specific edge case which disproves this rule?

You only need to be disproven once, for this to be disproven.

From Hypothesis to Experiment: Example

Simply disproving a hypothesis with a single example (in computer science — maths is a bit different) isn't all that useful

- Are only saying “given a single instance this claim is incorrect”
- Much more useful to discuss the performance of the algorithms average performance relative to each other
 - This is where generalisability of a result becomes important (more on this in the next live lecture)

From Hypothesis to Experiment: Example

An Improved Hypothesis: “GlasgowGPT is more factually accurate (on average) Claude, ChatGPT and Grok when answering factual questions”

Which we can turn into a research question:

- “Which LLM gives more factually accurate answers (on average): GlasgowGPT, Claude, ChatGPT and Grok?”
- Note: Often in computer science research questions are easier to address than hypothesis

More Examples of Hypothesis

H1: If I eat 500 less kcal per day, I will lose weight.

- Is **testable** (by weighing myself); is **falsifiable** (I might not lose weight); makes a **prediction** (reduced bodyweight); attempts to **explain an observation** (I'm not losing weight because I eat too much).

H2: If I drink coffee before going to lectures, I'll take more notes.

- Is **testable** (quantity of notes); is **falsifiable** (I might still fall asleep); makes a **prediction** (increased note-taking); attempts to **explain an observation** (I don't take much notes because I'm tired).

H3: If I turn the thermostat down, the gas bill will be lower.

- Is **testable** (compare gas use); is **falsifiable** (gas use might be similar); makes a **prediction** (cooler temperature uses less gas); attempts to **explain an observation** (bill is high because the thermostat is too high).

H4: If I practice reading Japanese for an extra 15 minutes per day and wear red socks, my Japanese reading level will improve.

- Is **testable** (reading level); is **falsifiable** (reading level might not improve); makes a **prediction** (higher reading level); attempts to **explain an observation** (my reading level isn't improving because I don't practice).
- But... is **not parsimonious** (socks are probably an irrelevant addition).

Ethics

Ethics

“moral principles that govern a person's behaviour or the conducting of an activity” (Oxford English Dictionary)

Acting ethically means acting in a way that embodies what people would consider as good values;

- We use ethical judgement when we reason about what actions to take
- In research, it's important to use similar reasoning about how to act
- Ethical treatises have been proposed for a long time — e.g. Aristotle, over 2000 years ago.

Why do we need ethics?

We must as researchers act responsible

As scientists we have an ethical / moral obligation to be responsible with the work that we do

- This, in turn, manifests itself as requirement of the job
 - E.g. university / companies have ethical systems in place to ensure any research conducted at them is done so to some ethical standard
- We could do science without ethics, but this is not normally publishable, and also limits us to what we can do and the morality of what we do
 - However, as scientists it is important to remember that ethics varies greatly around the world, and *not everything* gets ethical approval

We must as researchers act responsible

This need to conduct science ethically is due to repeated instances in the past where no regard was given for ethics

- There are numerous instances of these historically (we'll highlight a few of them to motivate why we need ethics system in place)
- The cynical view of this is such unethical science experiments continues to go on (it is just going on in secret)

Examples of What Not to Do